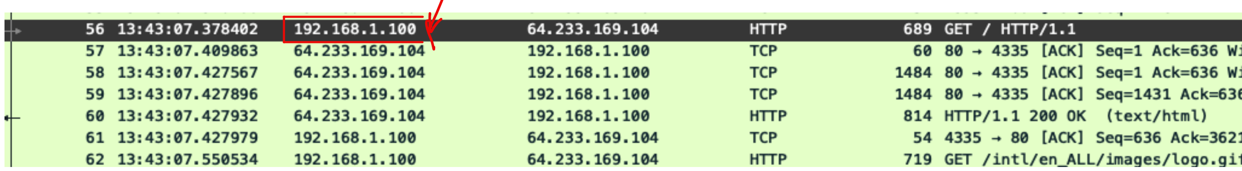
Freya Zou

Lab 06 NAT

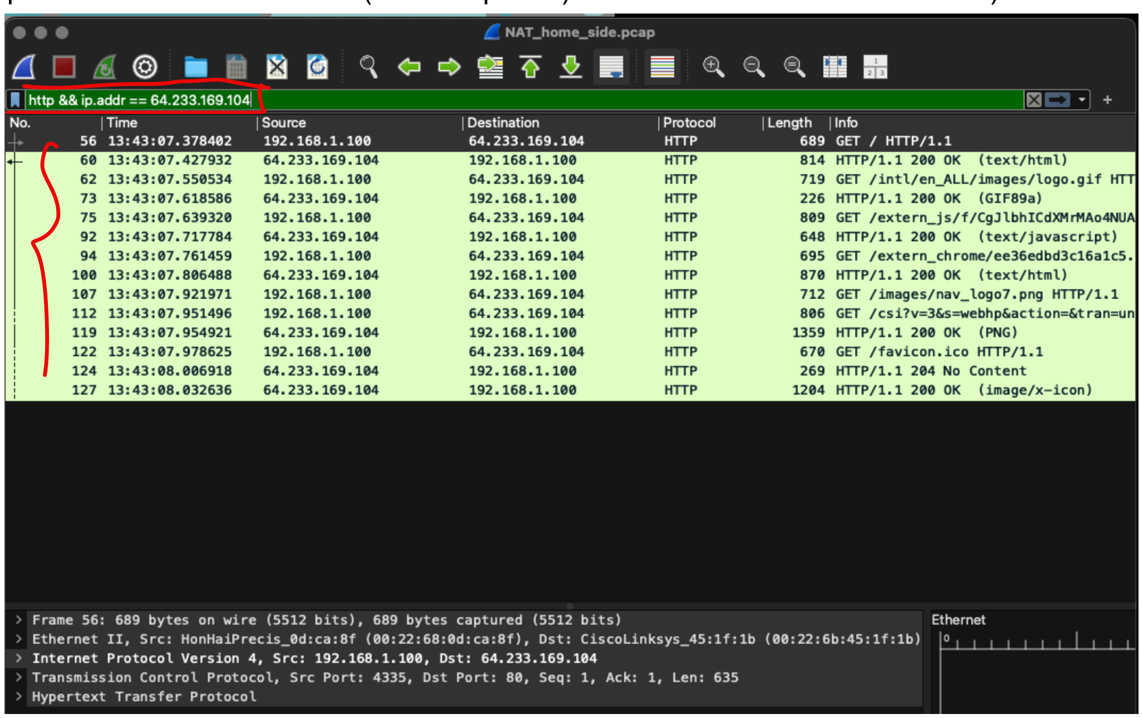
Note: I used the trace file in the lab Doc and didnt do extra credit.

1. NAT Measurement Scenario

1. What is the IP address of the client?
   The client address is 192.168.1.100

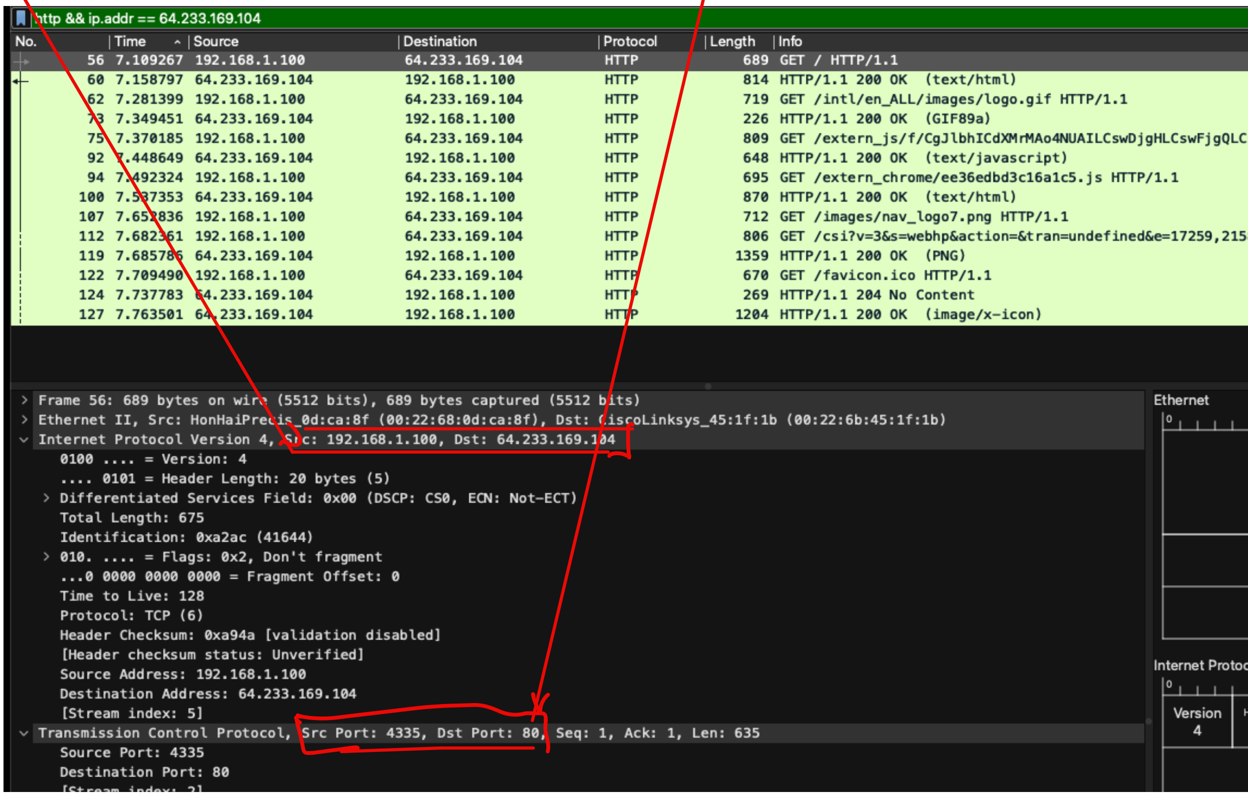


2. The client actually communicates with several different Google servers in order to implement "safe browsing." (See extra credit section at the end of this lab). The main Google server that will serve up the main Google web page has IP address 64.233.169.104. In order to display only those frames containing HTTP messages that are sent to/from this Google, server, enter the expression "http && ip.addr == 64.233.169.104" (without quotes) into the Filter: field in Wireshark).



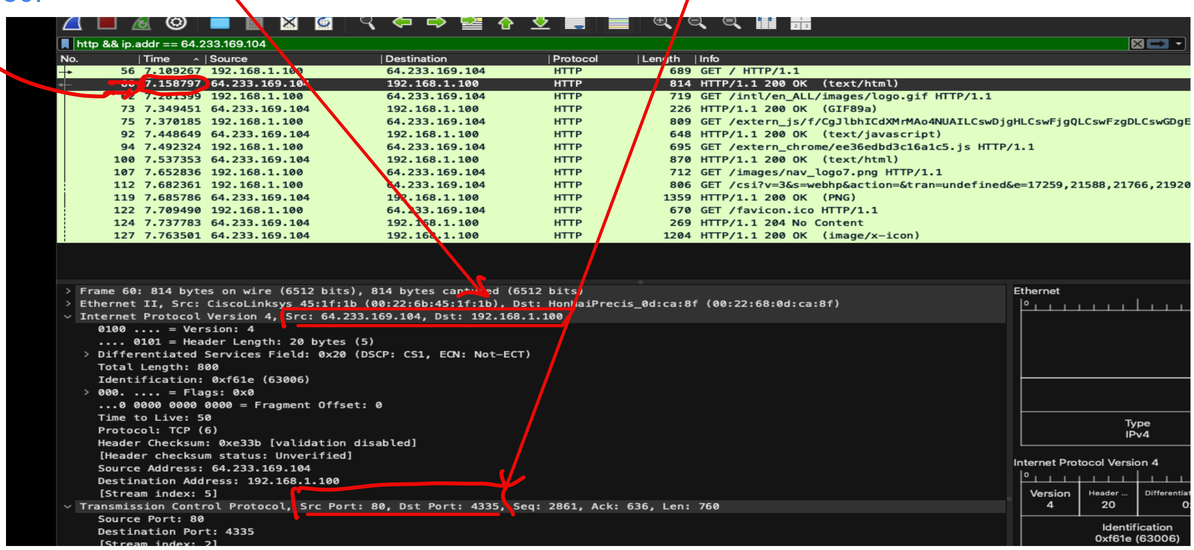
3. Consider now the HTTP GET sent from the client to the Google server (whose IP address is IP address 64.233.169.104) at time 7.109267. What are the source and destination IP addresses and TCP source and destination ports on the IP datagram carrying this HTTP GET?

The Source IP address is: 192.168.1.100; The destination IP address is: 64.233.164.104; The TCP source port is: 4335; The destination port is: 80



4. At what time^4 is the corresponding 200 OK HTTP message received from the Google server? What are the source and destination IP addresses and TCP source and destination ports on the IP datagram carrying this HTTP 200 OK message?

At the time 7.158797 is the corresponding 200 Ok HTTP message received from the Google server. The source IP address 64.233.169.104; the destination IP address is 192.168.1.100; The destination TCP port is: 4335; the source port is 80.



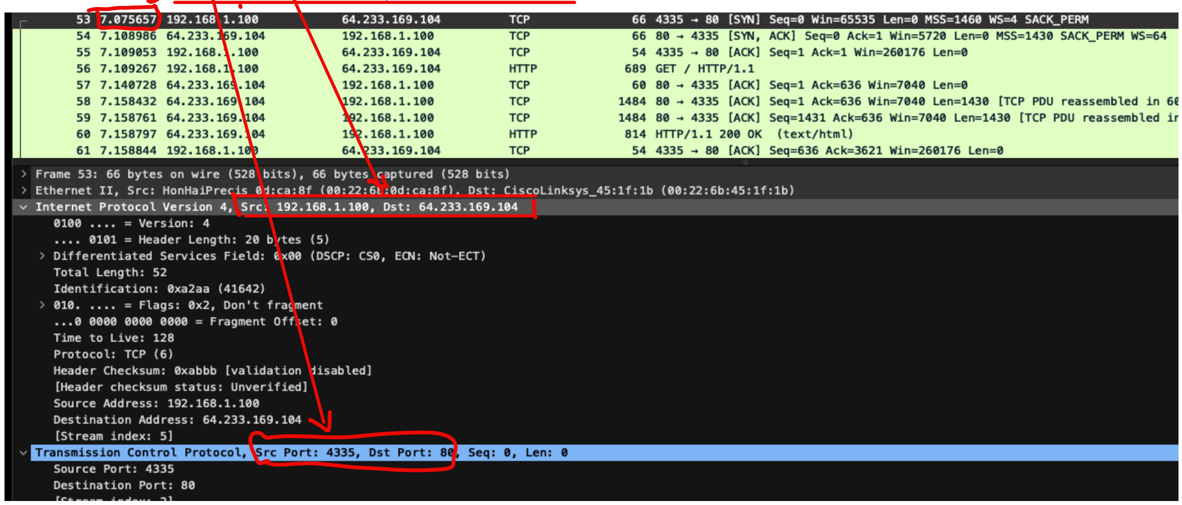
5. Recall that before a GET command can be sent to an HTTP server, TCP must first set up a connection using the three-way SYN/ACK handshake. At what time is the client-to-server TCP SYN segment sent that sets up the connection used by the GET sent at time 7.109267? What are the source and destination IP addresses and source and destination ports for the TCP SYN segment? What are the source and destination IP addresses and source and destination ports of the ACK sent in response to the SYN. At what time is this ACK received at the client? (Note: to find these segments you will need to clear the Filter expression you entered above in step 2. If you enter the filter "tcp", only TCP segments will be displayed by Wireshark).

SYN:

At the time 7.075657, this message appears in the trace file.

The source IP Src: 192.168.1.100, Dst: 64.233.169.104;

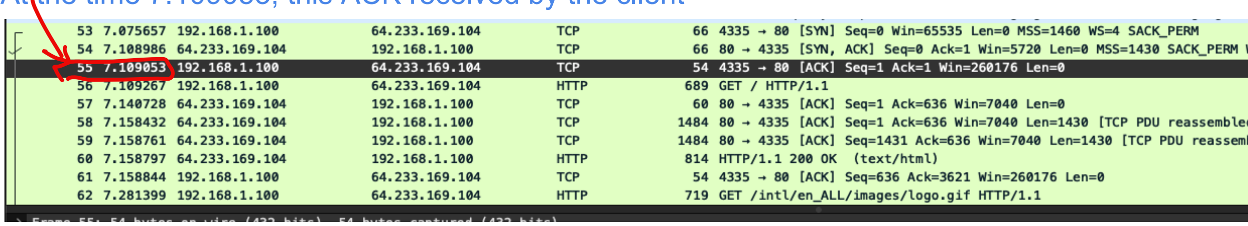
TCP: Src Port: 4335, Dst Port: 80


```
53 7.075657 192.168.1.100    64.233.169.104  TCP    66 4335 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=4 SACK_PERM
54 7.108986 64.233.169.104   192.168.1.100   TCP    66 80 → 4335 [SYN, ACK] Seq=0 Ack=1 Win=5720 Len=0 MSS=1430 SACK_PERM WS=64
55 7.109053 192.168.1.100    64.233.169.104  TCP    54 4335 → 80 [ACK] Seq=1 Ack=1 Win=260176 Len=0
56 7.109267 192.168.1.100    64.233.169.104  HTTP  689 GET / HTTP/1.1
57 7.140728 64.233.169.104   192.168.1.100   TCP    60 80 → 4335 [ACK] Seq=1 Ack=636 Win=7040 Len=0
58 7.158432 64.233.169.104   192.168.1.100   TCP  1484 80 → 4335 [ACK] Seq=1 Ack=636 Win=7040 Len=1430 [TCP PDU reassembled in 60
59 7.158761 64.233.169.104   192.168.1.100   TCP  1484 80 → 4335 [ACK] Seq=1431 Ack=636 Win=7040 Len=1430 [TCP PDU reassembled in
60 7.158797 64.233.169.104   192.168.1.100   HTTP  814 HTTP/1.1 200 OK  (text/html)
61 7.158844 192.168.1.100    64.233.169.104  TCP    54 4335 → 80 [ACK] Seq=636 Ack=3621 Win=260176 Len=0

> Frame 53: 66 bytes on wire (528 bits), 66 bytes captured (528 bits)
> Ethernet II, Src: HonHaiPrecis_0d:ca:8f (00:22:68:0d:ca:8f), Dst: CiscoLinksys_45:1f:1b (00:22:6b:45:1f:1b)
v Internet Protocol Version 4, Src: 192.168.1.100, Dst: 64.233.169.104
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 52
    Identification: 0xa2aa (41642)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0xabbb [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.100
    Destination Address: 64.233.169.104
    [Stream index: 5]
v Transmission Control Protocol, Src Port: 4335, Dst Port: 80, Seq: 0, Len: 0
    Source Port: 4335
    Destination Port: 80
```

ACK:

At the time 7.109053, this ACK received by the client

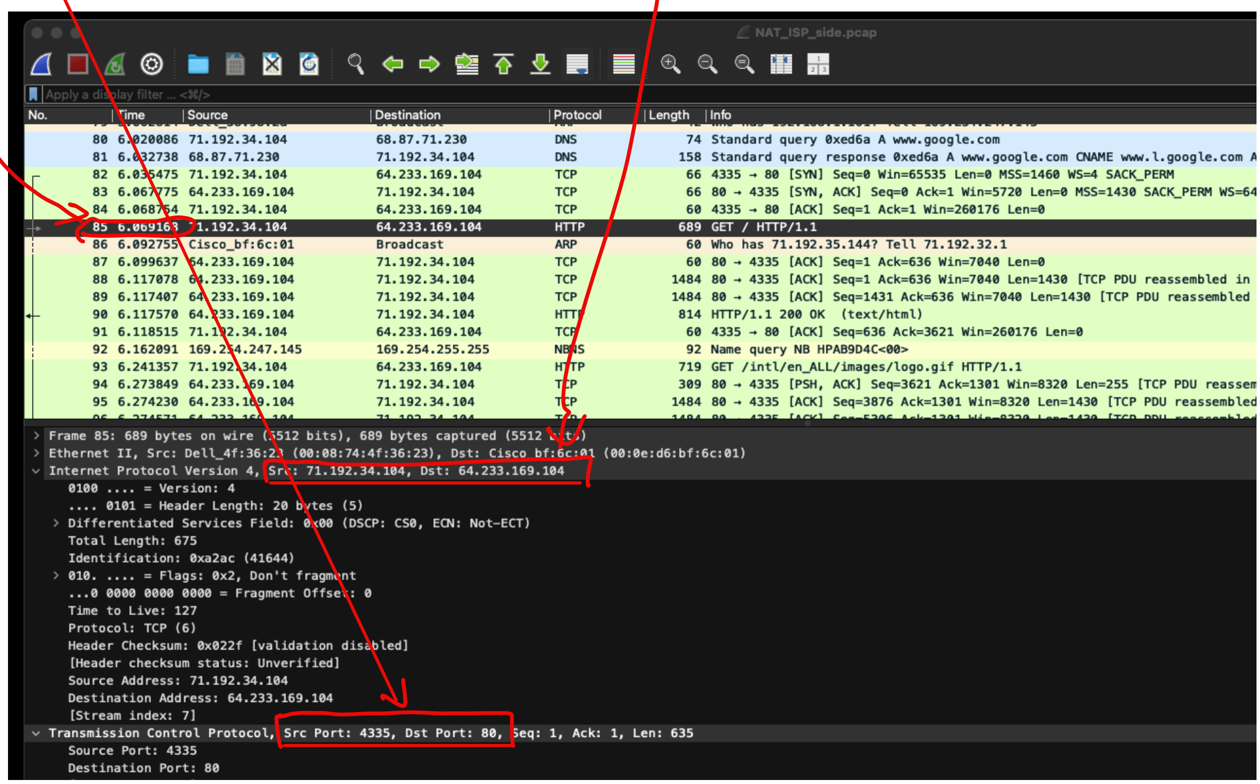

```
53 7.075657 192.168.1.100    64.233.169.104  TCP    66 4335 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=4 SACK_PERM
54 7.108986 64.233.169.104   192.168.1.100   TCP    66 80 → 4335 [SYN, ACK] Seq=0 Ack=1 Win=5720 Len=0 MSS=1430 SACK_PERM W
55 7.109053 192.168.1.100    64.233.169.104  TCP    54 4335 → 80 [ACK] Seq=1 Ack=1 Win=260176 Len=0
56 7.109267 192.168.1.100    64.233.169.104  HTTP  689 GET / HTTP/1.1
57 7.140728 64.233.169.104   192.168.1.100   TCP    60 80 → 4335 [ACK] Seq=1 Ack=636 Win=7040 Len=0
58 7.158432 64.233.169.104   192.168.1.100   TCP  1484 80 → 4335 [ACK] Seq=1 Ack=636 Win=7040 Len=1430 [TCP PDU reassemble
59 7.158761 64.233.169.104   192.168.1.100   TCP  1484 80 → 4335 [ACK] Seq=1431 Ack=636 Win=7040 Len=1430 [TCP PDU reassem
60 7.158797 64.233.169.104   192.168.1.100   HTTP  814 HTTP/1.1 200 OK  (text/html)
61 7.158844 192.168.1.100    64.233.169.104  TCP    54 4335 → 80 [ACK] Seq=636 Ack=3621 Win=260176 Len=0
62 7.281399 192.168.1.100    64.233.169.104  HTTP  719 GET /intl/en_ALL/images/logo.gif HTTP/1.1
```

IP Source: 64.233.169.104 Dest: 192.168.1.100

TCP: Source: 80 Dst: 4335

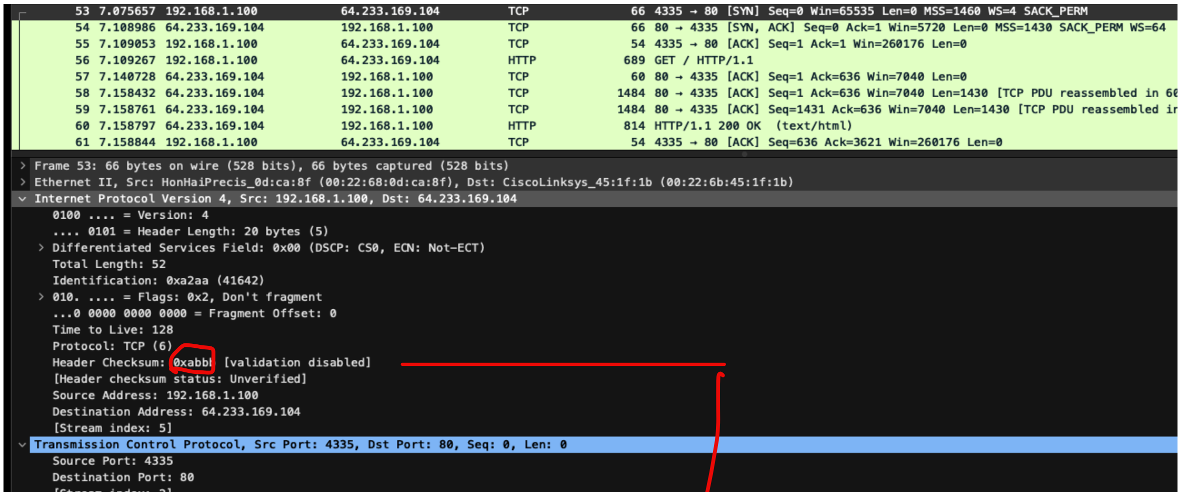
6. In the NAT_ISP_side trace file, find the HTTP GET message was sent from the client to the Google server at time 7.109267 (where t=7.109267 is time at which this was sent as recorded in the NAT_home_side trace file). At what time does this message appear in the NAT_ISP_side trace file? What are the source and destination IP addresses and TCP source and destination ports on the IP datagram carrying this HTTP GET (as recording in the NAT_ISP_side trace file)? Which of these fields are the same, and which are different, than in your answer to question 3 above?

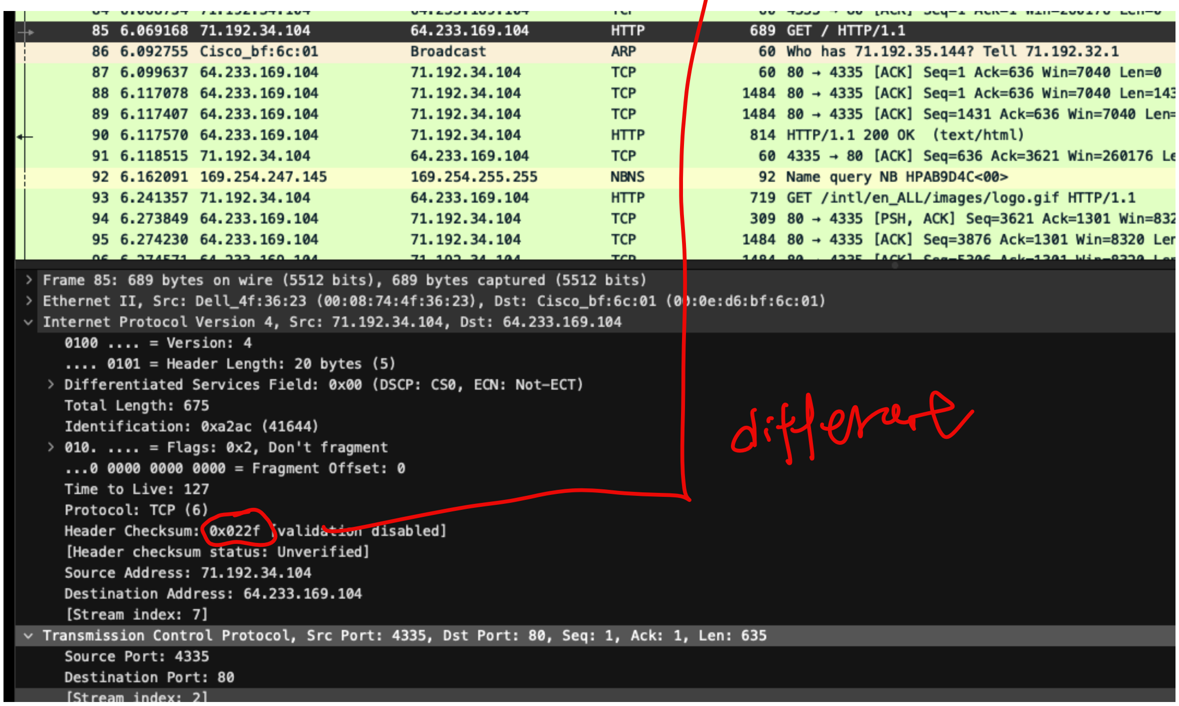
At the same time, 6.069168 this message appeared in the trace file,
IP: Src: 71.192.34.104, Dst: 64.233.169.104
TCP: Src Port: 4335, Dst Port: 80
The Src port, the Dst port number, and the Dest IP address are the same. Only the source IP address is different.



7. Are any fields in the HTTP GET message changed? Which of the following fields in the IP datagram carrying the HTTP GET are changed: Version, Header Length,Flags, Checksum. If any of these fields have changed, give a reason (in one sentence) stating why this field needed to change.

There are no fields in the HTTP GEt message changed. For Version, headerLength, and Flags are the same, which means no changes, the only field changed is the header checksum. The header checksum is changed because the source IP address is changed.





8. In the NAT_ISP_side trace file, at what time is the first 200 OK HTTP message received from the Google server? What are the source and destination IP addresses and TCP source and destination ports on the IP datagram carrying this HTTP 200 OK message? Which of these fields are the same, and which are different than your answer to question 4 above?

At the time 6.117570

IP: Src: 64.233.169.104, Dst: 71.192.34.104

TCP in IP datagram: Src Port: 80, Dst Port: 4335

The time and the destination IP address are different; The source IP address, the TCP source port, and the destination port are the same.

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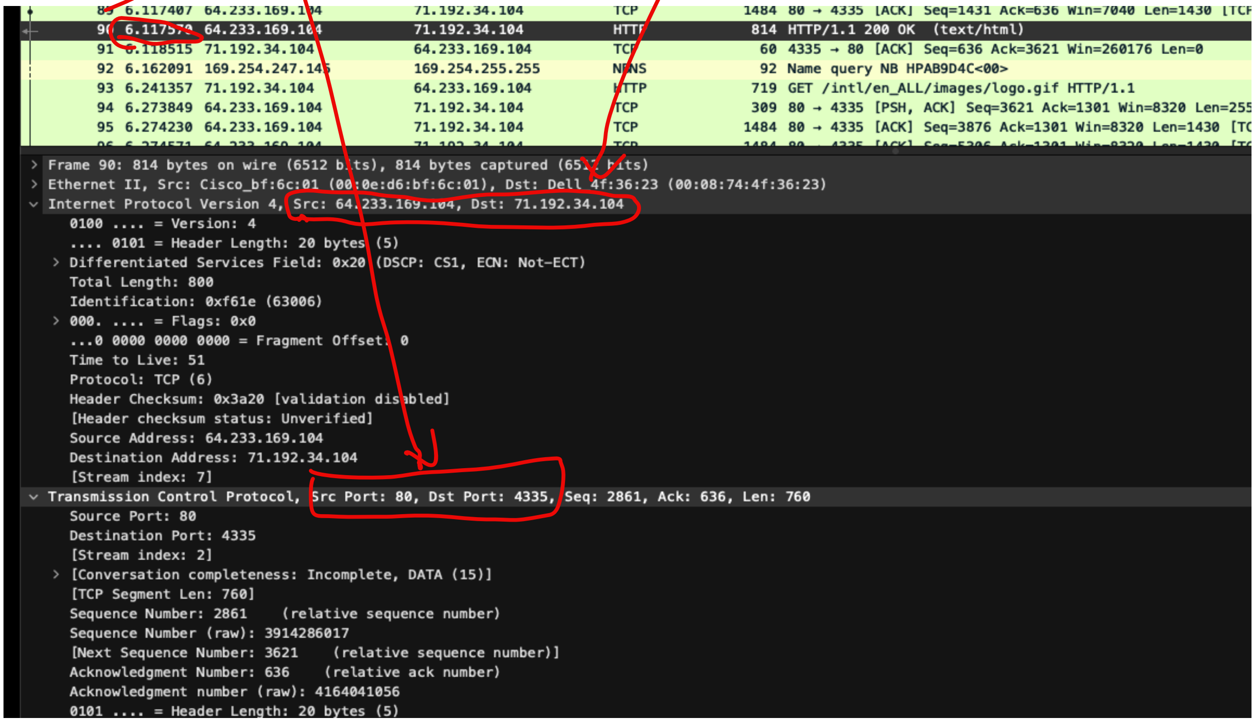
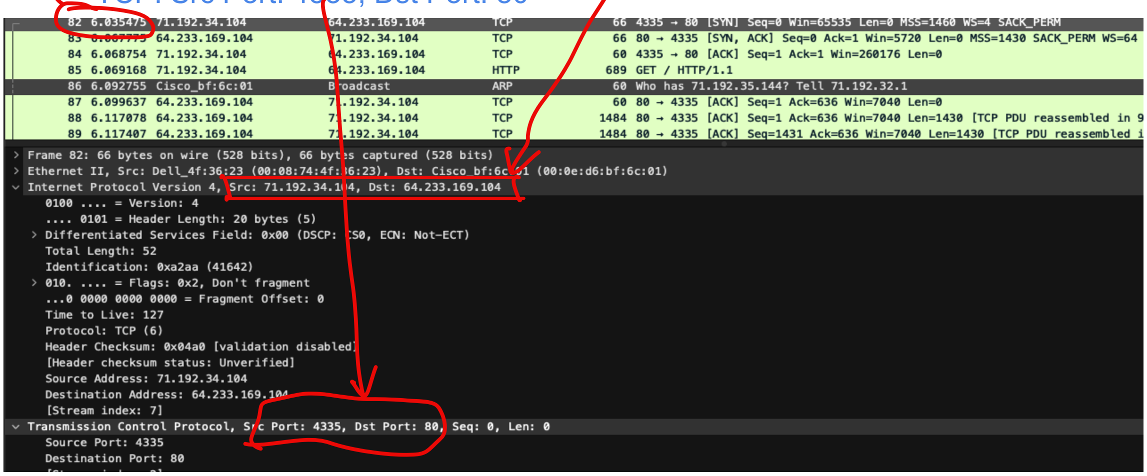


9. In the NAT_ISP_side trace file, at what time were the client-to-server TCP SYN segment and the server-to-client TCP ACK segment corresponding to the segments in question 5 above captured? What are the source and destination IP addresses and source and destination ports for these two segments? Which of these fields are the same, and which are different than your answer to question 5 above?

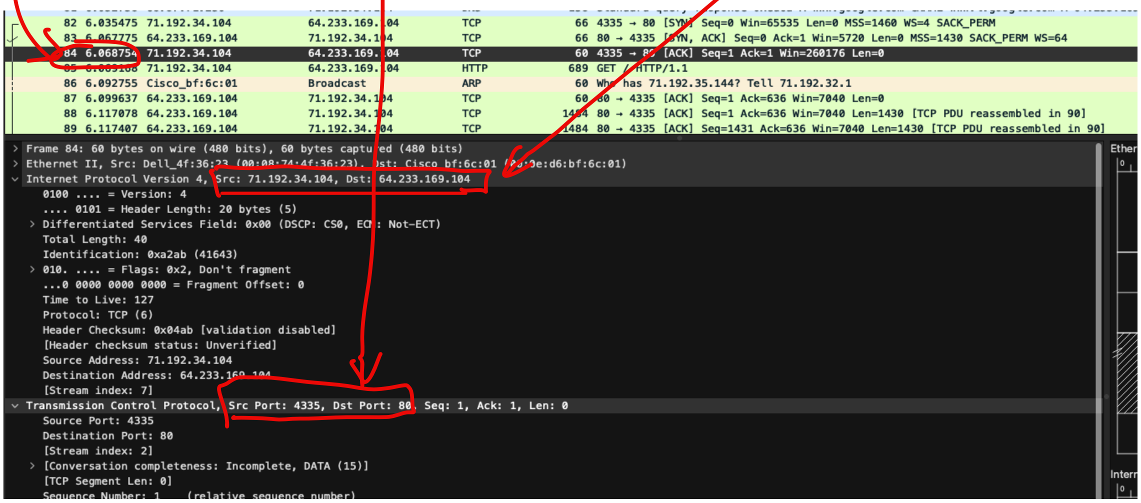
SYN:

At the time 6.035475
IP: Src: 71.192.34.104, Dst: 64.233.169.104
TCP: Src Port: 4335, Dst Port: 80



ACK:

At the time 6.068754
IP address: Src: 71.192.34.104, Dst: 64.233.169.104
TCP: Src Port: 4335, Dst Port: 80



For the SYN packet, only the source IP address is different, and the destination IP address, the TCP destination port, and source port number are the same.

For the ACK packet, only the source IP address has changed, and the destination IP address, the TCP destination and source port number are the same.

Thus, only the source IP addresses are different, the rest of it is the same.